

Appendix for: The New Keynesian Cross

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Abstract

This Appendix outlines derivations accompanying "The New Keynesian Cross".

A RANK Derivations

An agent j chooses consumption, asset holdings and leisure solving the standard intertemporal problem: $\max E_0 \sum_{t=0}^{\infty} \beta^t U(C_t^j, N_t^j)$, subject to the sequence of constraints:

$$B_t^j + \Omega_{t+1}^j V_t \leq Z_t^j + \Omega_t^j (V_t + P_t D_t) + W_t N_t^j - P_t C_t^j.$$

C_t^j, N_t^j are consumption and hours worked, B_t^j is the nominal value at *end of period* t of a portfolio of all state-contingent assets held, except for shares in firms—likewise for Z_t^j , beginning of period wealth.¹ V_t is average market value at time t of shares, D_t their real dividend payoff and Ω_t^j are share holdings. Absence of arbitrage implies that there exists a stochastic discount factor $Q_{t,t+1}^j$ such that the price at t of a portfolio with uncertain payoff at $t+1$ is (for state-contingent assets and shares respectively, for an agent j who participates in those markets):

$$\frac{B_{j,t}}{P_t} = E_t \left[Q_{t,t+1}^j \frac{Z_{j,t+1}}{P_{t+1}} \right] \text{ and } \frac{V_t}{P_t} = E_t \left[Q_{t,t+1}^j \left(\frac{V_{t+1}}{P_{t+1}} + D_{t+1} \right) \right], \quad (1)$$

which iterated forward gives the fundamental pricing equation: $\frac{V_t}{P_t} = E_t \sum_{i=1}^{\infty} Q_{t,t+i}^j D_{t+i}$. The

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¹We distinguish shares from the other assets explicitly since their distribution plays a crucial role in the rest of the analysis.

riskless gross short-term REAL interest rate R_t is a solution to:

$$\frac{1}{R_t} = E_t Q_{t,t+1}^j \quad (2)$$

Note that for nominal assets we have the nominal interest rate $\frac{1}{I_t} = E_t \frac{P_t}{P_{t+1}} Q_{t,t+1}^j$.

Substituting the no-arbitrage conditions (1) into the wealth dynamics equation gives the flow budget constraint. Together with the usual 'natural' no-borrowing limit for *each* state, and anticipating that in equilibrium all agents will hold a constant fraction of the shares (there is no trade in shares) Ω^j (whose integral across agents is 1), this implies the usual intertemporal budget constraint:

$$\begin{aligned} E_t \left[\frac{P_t}{P_{t+1}} Q_{t,t+1}^j X_{t+1}^j \right] &\leq X_t^j + W_t N_t^j - P_t C_t^j. \\ X_t^j &= Z_t^j + \Omega^j (V_t + P_t D_t) \\ &= Z_t^j + \Omega^j \left(E_t \sum_{i=0}^{\infty} P_t Q_{t,t+i}^j D_{t+i} \right) \\ E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j C_{t+i}^j &\leq \frac{X_t^j}{P_t} + E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j \frac{W_{t+i}}{P_{t+i}} N_{t+i}^j \\ &= E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j Y_{t+i}^j \end{aligned} \quad (3)$$

where

$$Y_{t+i}^j = \Omega^j D_{t+i} + \frac{W_{t+i}}{P_{t+i}} N_{t+i}^j \quad (4)$$

is income of agent j . Maximizing utility subject to this constraint gives the first-order necessary and sufficient conditions at each date and in each state:

$$\beta \frac{U_C(C_{t+1}^j)}{U_C(C_t^j)} = Q_{t,t+1}^j$$

along with (3) holding with equality (or alternatively flow budget constraint holding with equality and transversality conditions ruling out Ponzi games be satisfied: $\lim_{i \rightarrow \infty} E_t [Q_{t,t+i}^j Z_{t+i}^j] = \lim_{i \rightarrow \infty} E_t [Q_{t,t+i}^j V_{t+i}] = 0$). Using (3) and the functional form of the utility function the short-

term nominal interest rate must obey:

$$\frac{1}{R_t} = \beta E_t \left[\frac{U_C(C_{t+1}^j)}{U_C(C_t^j)} \right].$$

Denote by small letter log deviations from steady-state, except for rates of return (where they denote absolute deviations). Notice that

$$Q_{t,t+i} = \beta^i \frac{U_C(C_{t+i}^j)}{U_C(C_t^j)}$$

and in steady state: $Q_i = \beta^i$. Thus we have

$$q_{t,t+i}^j = \ln \frac{Q_{t,t+i}^j}{Q_i^j} = \ln \frac{U_C(C_{t+i}^j)}{U_C(C_t^j)} = -\sigma^{-1} (c_{t+i}^j - c_t^j),$$

where

$$q_{t,t+i}^j = q_{t,t+1}^j + q_{t+1,t+2}^j + \dots + q_{t+i-1,t+i}^j$$

The Euler equation is:

$$r_t = -E_t q_{t,t+1}^j$$

Rewrite as

$$c_t^j = E_t c_{t+1}^j - \sigma r_t$$

and iterate forward, using $q_{t,t+i}^j = -\sum_{k=0}^{i-1} r_{t+k}$

$$c_t^j = E_t c_{t+i}^j + \sigma E_t q_{t,t+i}^j \tag{5}$$

Now loglinearize intertemporal budget constraint

$$E_t \sum_{i=0}^{\infty} \beta^i (q_{t,t+i}^j + c_{t+i}^j) = E_t \sum_{i=0}^{\infty} \beta^i (q_{t,t+i}^j + y_{t+i}^j)$$

and add to each side $(\sigma - 1) \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^j$

$$E_t \sum_{i=0}^{\infty} \beta^i (\sigma q_{t,t+i}^j + c_{t+i}^j) = E_t \sum_{i=0}^{\infty} \beta^i (\sigma q_{t,t+i}^j + y_{t+i}^j)$$

By virtue of the Euler equation the LHS simplifies

$$\frac{1}{1-\beta}c_t^j = \sigma \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^j + \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^j$$

Develop RHS, use $q_{t,t}^j = 0$:

$$\begin{aligned} \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^j &= 0 - \sum_{i=1}^{\infty} \beta^i E_t \sum_{k=0}^{i-1} r_{t+k} \\ &= -\frac{\beta}{1-\beta} \sum_{i=0}^{\infty} \beta^i E_t r_{t+i} \end{aligned}$$

And replace to obtain (multiplying by $1-\beta$)

$$\begin{aligned} c_t^j &= -\sigma\beta \sum_{i=0}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^j \\ &= -\sigma\beta r_t + (1-\beta) y_t^j - \sigma\beta \sum_{i=1}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=1}^{\infty} \beta^i E_t y_{t+i}^j \end{aligned}$$

Now replace the expression for expected consumption tomorrow

$$\beta c_{t+1}^j = -\sigma\beta \sum_{i=0}^{\infty} \beta^{i+1} E_t r_{t+1+i} + (1-\beta) \sum_{i=0}^{\infty} \beta^{i+1} E_t y_{t+1+i}^j$$

to obtain the consumption function in text:

$$c_t^j = (1-\beta) y_t^j - \sigma\beta r_t + \beta E_t c_{t+1}^j. \quad (6)$$

B TANK Derivations

Consider the TANK model without the assumption of optimal steady-state subsidy, so with positive profits and lack of steady-state consumption insurance. In particular, instead of assuming $\tau^S = (\varepsilon - 1)^{-1}$ we set it free: using the steady-state pricing condition $\frac{WN}{PC} = 1 - \Phi$ where I defined the steady-state distortion following Woodford (2003), Chapter 6 as $\Phi \equiv 1 - \frac{(1+\tau^S)(\varepsilon-1)}{\varepsilon} \in [0; 1]$. A value of 0 corresponds to an undistorted steady-state equilibrium (such as under an optimal subsidy $\tau^S = (\varepsilon - 1)^{-1}$) and the maximum of 1 obtains when the markup tends to infinity and the subsidy s is fixed.² Using this, the profit share in

²Other distortions such as imperfect labor markets and a wage markup, or distortionary taxation would change the exact expression for Φ but neither its interpretation nor the limit result that under appropriately

consumption in steady state is: $\frac{D}{C} = 1 - \frac{(W/P)N}{C} = \Phi$, and the transfer share $\frac{T_H}{C} = \frac{\tau^D}{\lambda} \frac{D}{C}$, delivering the share of H consumption in total:

$$\frac{C_H}{C} = \frac{(W/P)N}{C} + \frac{Transfer^H}{C} = 1 - \left(1 - \frac{\tau^D}{\lambda}\right) \Phi$$

Approximating around this steady state we have profit deviations as share of C , $d_t = (D_t - D)/C$

$$d_t = c_t - (1 - \Phi)(w_t + n_t) = \Phi c_t - (1 - \Phi)w_t.$$

The approximation of transfers is still $\frac{\tau^D}{\lambda} d_t$ with the wage schedule (aggregating labor supplies and using goods market clearing and production function) $w_t = (\varphi + \gamma) c_t$. Replacing all this in the approximation of H 's budget constraint:

$$\frac{C_H}{C} c_{H,t} = \frac{(W/P)N}{C} (w_t + n_{H,t}) + \frac{\tau^D}{\lambda} d_t$$

we obtain their consumption function:

$$c_{H,t} = \chi(\Phi) y_t, \\ \text{where } \chi(\Phi) \equiv 1 + \varphi \left(1 - \frac{\tau^D}{\lambda}\right) \Upsilon \text{ with } \Upsilon \equiv \frac{1}{1 + \frac{\tau^D}{\lambda} \frac{\Phi}{1-\Phi} \frac{\varphi}{\varphi+\gamma}} \in [0; 1]$$

which shows that our main conclusions and intuition for the benchmark TANK model in text carry through to the case of more general fiscal policy, without full steady-state redistribution and without optimal subsidy.³

Consider now the case of **decreasing returns** and of a different **labor market** setup, used in Debortoli and Galí (2018), and previously by Galí, Lopez-Salido and Valles (2007): a centralized labor market pooling everybody's labor input and supplying demand-determined labor according to an aggregate wage schedule; the production function is $Y = N^\zeta$ and the individual labor supply optimality conditions are replaced by a wage schedule and a condition stating that everybody works the same hours, in loglinearized form $\varphi n_t = w_t - \sigma^{-1} c_t$ and $n_t^H = n_t^S = n_t$. With this setup the profit function becomes $d_t = c_t - \zeta(1 - \Phi)(w_t + n_t)$ and

designed taxes it can be eliminated—so Φ can be regarded as a general index of supply distortions.

³The more general point about redistribution is as follows: given an income function for H, say $C_t^H = \Gamma(Y_t) + \mathcal{T}$, a transfer reduces the elasticity of their after-tax income to aggregate income. Letting $\chi^\tau = \frac{\Gamma_Y Y}{\Gamma + \mathcal{T}}$, it follows immediately that as long as $\mathcal{T} > 0$ we have $\chi^\tau < \chi^0$ and if it is high enough, $\chi^\tau < 1 < \chi^0$, where χ^0 is the elasticity under zero transfer.

the consumption function of H (replacing everything):

$$\begin{aligned} c_{H,t} &= \tilde{\chi} y_t \\ \tilde{\chi} &= 1 + \left(1 - \frac{\tau^D}{\lambda}\right) \left(\frac{1 - \zeta + \varphi}{\zeta} + \gamma\right) \tilde{\Upsilon} \text{ with } \tilde{\Upsilon} = \frac{1}{1 + \frac{\tau^D}{\lambda} \left(\frac{1}{\zeta(1-\Phi)} - 1\right)} \end{aligned}$$

The main insight regarding χ carries through: whether it is smaller or larger than 1 depends on the progressivity of taxes, in particular whether τ^D is smaller or larger than λ . Other than that, the main difference is that χ (when > 1) is larger ceteris paribus under DRS $\zeta < 1$, and it is larger under a centralized labor market even with CRS and optimal subsidy $(1 + \left(1 - \frac{\tau^D}{\lambda}\right)(\varphi + \gamma))$; for instance, with no redistribution with competitive labor market it is 2, and with centralized labor market it is 3.

C Analytical HANK Derivations

The loglinearized self-insurance Euler equation of S around the symmetric steady-state is:

$$c_t^S = s E_t c_{t+1}^S + (1 - s) E_t c_{t+1}^H - \sigma r_t.$$

Noticing that we have, as before in TANK, whatever the redistribution scheme determining $\chi \equiv 1 + \varphi \left(1 - \frac{\tau^D}{\lambda}\right) \leq 1$:

$$c_t^H = y_t^H = \chi y_t$$

we obtain the aggregate Euler-IS for my model:

$$\begin{aligned} c_t &= \delta E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} r_t, \\ \text{where } \delta &\equiv 1 + (\chi - 1) \frac{1 - s}{1 - \lambda \chi}. \end{aligned} \tag{7}$$

Using the stochastic discount factor notation, we now have

$$\sigma q_{t,t+1}^S = c_t^S - s E_t c_{t+1}^S - (1 - s) E_t c_{t+1}^H$$

Iterating forward (note: we no longer have $q_{t,t+i}^j = -\sum_{k=0}^{i-1} r_{t+k}$)

$$c_t^S = s^i E_t c_{t+i}^S - \sigma \sum_{k=0}^{i-1} s^k (r_{t+k} - (1-s) E_t c_{t+k}^H) \quad (8)$$

$$c_t^S = s^i E_t c_{t+i}^S + \sigma E_t \sum_{k=0}^{i-1} s^k (q_{t,t+k}^S + (1-s) E_t c_{t+k}^H) \quad (9)$$

Using the definition of stochastic discount factor:

$$\begin{aligned} \sigma q_{t,t+i}^S &= c_t^S - s E_t c_{t+1}^S - (1-s) E_t c_{t+1}^H + c_{t+1}^S - s E_t c_{t+2}^S - (1-s) E_t c_{t+2}^H + \\ &\quad \dots + c_{t+i-1}^S - s E_t c_{t+i}^S - (1-s) E_t c_{t+i}^H \\ \sigma q_{t,t+i}^S + c_{t+i}^S &= c_t^S + (1-s) E_t \sum_{k=1}^i (c_{t+k}^S - c_{t+k}^H) \end{aligned}$$

Now loglinearize the intertemporal budget constraint

$$E_t \sum_{i=0}^{\infty} \beta^i (q_{t,t+i}^S + c_{t+i}^S) = E_t \sum_{i=0}^{\infty} \beta^i (q_{t,t+i}^S + y_{t+i}^S)$$

Add to each side $(\sigma - 1) \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^S$

$$E_t \sum_{i=0}^{\infty} \beta^i (\sigma q_{t,t+i}^S + c_{t+i}^S) = E_t \sum_{i=0}^{\infty} \beta^i (\sigma q_{t,t+i}^S + y_{t+i}^S)$$

By virtue of the Euler equation the LHS simplifies

$$\begin{aligned} \frac{1}{1-\beta} c_t^S + (1-s) E_t \sum_{i=0}^{\infty} \beta^i \sum_{k=1}^i (c_{t+k}^S - c_{t+k}^H) &= \sigma \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^S + \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^S \\ \frac{1}{1-\beta} c_t^S + \frac{1-s}{1-\beta} E_t \sum_{i=1}^{\infty} \beta^i (c_{t+i}^S - c_{t+i}^H) &= \sigma \sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^S + \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^S \end{aligned}$$

Develop RHS $\sum_{i=0}^{\infty} \beta^i E_t q_{t,t+i}^e$ using $q_{t,t} = 0$, this is as above in general case and replace to

obtain (multiplying by $1 - \beta$)

$$\begin{aligned} c_t^S &= -(1-s) E_t \sum_{i=1}^{\infty} \beta^i (c_{t+i}^S - c_{t+i}^H) - \sigma \beta \sum_{i=0}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^S \\ &= -\sigma \beta r_t + (1-\beta) y_t^S - (1-s) E_t \sum_{i=1}^{\infty} \beta^i (c_{t+i}^S - c_{t+i}^H) - \sigma \beta \sum_{i=1}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=1}^{\infty} \beta^i E_t y_{t+i}^S \end{aligned}$$

Now replace expression for expected consumption tomorrow

$$\beta c_{t+1}^S = -(1-s) E_t \sum_{i=1}^{\infty} \beta^{i+1} (c_{t+i+1}^S - c_{t+i+1}^H) - \sigma \beta \sum_{i=0}^{\infty} \beta^{i+1} E_t r_{t+1+i} + (1-\beta) \sum_{i=0}^{\infty} \beta^{i+1} E_t y_{t+1+i}^S$$

to obtain the consumption function:

$$c_t^S = -\sigma \beta r_t + (1-\beta) y_t^S - (1-s) E_t \sum_{i=1}^{\infty} \beta^i (c_{t+i}^S - c_{t+i}^H) - \sigma \beta \sum_{i=1}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=1}^{\infty} \beta^i E_t y_{t+i}^S$$

or in recursive form:

$$\begin{aligned} c_t^S &= -\sigma \beta r_t + (1-\beta) y_t^S - (1-s) \beta (E_t c_{t+1}^S - E_t c_{t+1}^H) + \beta E_t c_{t+1}^S \\ &= -\sigma \beta r_t + (1-\beta) y_t^S + \beta s E_t c_{t+1}^S + \beta (1-s) E_t c_{t+1}^H \end{aligned}$$

Aggregate and use $c_t^H = y_t^H = \chi y_t$ to obtain (using the notation for $\delta = \frac{s+(1-\lambda-s)\chi}{1-\lambda\chi}$)

$$c_t = [1 - \beta(1 - \lambda\chi)] y_t - (1 - \lambda) \sigma \beta r_t + \beta \delta (1 - \lambda\chi) E_t c_{t+1}.$$

To find the effects of FG we iterate forward:

$$c_t = -(1 - \lambda) \sigma \beta \sum_{i=0}^{\infty} [\beta \delta (1 - \lambda\chi)]^i E_t r_{t+i} + [1 - \beta(1 - \lambda\chi)] \sum_{i=0}^{\infty} [\beta \delta (1 - \lambda\chi)]^i E_t y_{t+i}$$